Quantum Mechanics II — Exercise Sheet 15

Hydrogen levels and their states

Variant: Solutions

Exercise 1 (Normalizing the second level). Put $a_0 = \hbar^2/(mC)$ and $r = |x|$. The level $-\mathrm{Ry}/4$ contains the real functions

$$f_0 = (1 - r/(2a_0))e^{-r/(2a_0)}, \qquad f_j = x_j e^{-r/(2a_0)}, \quad j = 1, 2, 3.$$

(i) Verify the $l = 0$ and $l = 1$ radial equations for these functions. (ii) Normalize all four with positive real constants and show that they are mutually orthogonal. You may use $\int_0^\infty r^n e^{-r/a_0}\,\mathrm{d}r = n!\,a_0^{n+1}$ for integers $n \geq 0$.

Solution. The radial functions are $u_0 = r(1 - r/(2a_0))e^{-r/(2a_0)}$ and $u_1 = r^2 e^{-r/(2a_0)}$. Their differentiated forms are

$$u_0'' = e^{-r/(2a_0)}\left(-\frac{2}{a_0} + \frac{5r}{4a_0^2} - \frac{r^2}{8a_0^3}\right), \qquad u_1'' = e^{-r/(2a_0)}\left(2 - \frac{2r}{a_0} + \frac{r^2}{4a_0^2}\right).$$

Substitution in $H_l = -\hbar^2 \mathrm{d}^2/(2m\mathrm{d}r^2) + \hbar^2 l(l+1)/(2mr^2) - C/r$ gives $H_l u_l = -\mathrm{Ry}\,u_l/4$ for $l = 0, 1$. The factorial integrals give

$$\|f_0\|^2 = 4\pi\,(2 - 6 + 6)\,a_0^3 = 8\pi a_0^3, \qquad \|f_j\|^2 = \frac{4\pi}{3}\,4!\,a_0^5 = 32\pi a_0^5.$$

The normalized functions are

$$\phi_0 = \frac{f_0}{\sqrt{8\pi a_0^3}}, \qquad \phi_j = \frac{f_j}{\sqrt{32\pi a_0^5}}, \quad j = 1, 2, 3.$$

For different Cartesian indices the cross integrand is odd in one coordinate. The cross term between f_0 and f_j is odd in x_j. All four functions are therefore orthonormal.

Exercise 2 (Resolving an angular superposition). For a fixed hydrogen principal number $n \geq 3$ let $|l, \mu\rangle$ denote normalized orbital eigenvectors at that level. Take $\psi = (|0,0\rangle + \sqrt{2}|1,0\rangle + \mathrm{i}|2,1\rangle)/2$. (i) Find the distributions of energy, L^2 and L_3. (ii) Find the vector after retaining $L_3 = 0$, and decide whether that reading fixes l.

Solution. The three coefficient probabilities are $1/4$, $1/2$ and $1/4$. They are all at the same energy, so the energy outcome is $-\mathrm{Ry}/n^2$ with probability one. The angular distributions are

L^2	0	$2\hbar^2$	$6\hbar^2$
probability	$1/4$	$1/2$	$1/4$

, | L_3 | 0 | $\hbar$ |
|---|---|---|
| probability | $3/4$ | $1/4$ |

.

The retained vector is

$$\psi_{\mathrm{kept}} = \frac{|0,0\rangle + \sqrt{2}|1,0\rangle}{\sqrt{3}}.$$

It still has two possible values of l, so the L_3 reading does not fix l.

Exercise 3 (A line series determines its limit). Neglect recoil and let $\nu_{n\to 2}$ be the photon frequency for hydrogen transitions from principal number $n > 2$ to 2. Find the ratio of the series-limit frequency to $\nu_{3\to 2}$. If the same Coulomb constant is used for a nucleus of mass M and an orbiting particle of mass m_e, state how all these frequencies scale with $m_{\mathrm{red}} = m_e M/(m_e + M)$. Distinguish a limiting frequency from a transition starting in a normalizable state at zero energy.

Solution. The photon energy is the difference of the two bound energies:

$$2\pi\hbar\nu_{n\to 2} = \mathrm{Ry}\left(\frac{1}{4} - \frac{1}{n^2}\right).$$

At the series limit the bracket tends to $1/4$, while for $n = 3$ it is $5/36$. Hence

$$\frac{\nu_{\infty\to 2}}{\nu_{3\to 2}} = \frac{9}{5}.$$

Since $\mathrm{Ry} \propto m_{\mathrm{red}}$, all frequencies scale by m_{red}/m_e relative to an infinitely heavy nucleus of the same charge. The limit is approached by bound states with increasing n; zero itself is not an additional normalizable hydrogen energy eigenstate.

Exercise 4 (The projected Stark matrix). Use the four normalized real vectors $\phi_0, \phi_1, \phi_2, \phi_3$ of Exercise 1, with their positive constants. Let Π_2 project onto their span and consider the finite matrix of $\Pi_2(Fx_3)\Pi_2$ for $F \in \mathbb{R}$. Compute all its entries and eigenvalues. Interpret the eigenvalues as the formal first-order level shifts only. No isolated eigenvalue or convergent perturbation series is assumed for the full Hamiltonian with the unbounded linear potential.

Solution. Parity makes every entry zero except the pair coupling ϕ_0 and ϕ_3. The remaining integral is

$$\langle \phi_0, x_3\phi_3\rangle = \frac{4\pi}{3\sqrt{8\pi a_0^3}\sqrt{32\pi a_0^5}} \int_0^\infty r^4\left(1 - \frac{r}{2a_0}\right) e^{-r/a_0}\,\mathrm{d}r = -3a_0.$$

In the ordered basis $(\phi_0, \phi_1, \phi_2, \phi_3)$ the projected matrix is

$$F\begin{pmatrix} 0 & 0 & 0 & -3a_0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -3a_0 & 0 & 0 & 0 \end{pmatrix}.$$

Its eigenvalues are $-3Fa_0$, 0, 0 and $3Fa_0$. The coupled eigenvectors are $(\phi_0 + \phi_3)/\sqrt{2}$ and $(\phi_0 - \phi_3)/\sqrt{2}$. This is an exact calculation of the projected matrix, and only a formal first-order statement about the full Stark problem.